

NANOSENSORS For Agricultural Pest Management



Dr Deepa Bhagat is a scientist at ICAR-National Bureau of Agricultural Insect Resources, Bangalore



Dr Parikshit Maitra is scientist-0 at Technical Research Centre, Indian Association for the Cultivation of Science, Kolkata

Globally, 20-40 per cent of crop yields are lost due to plant pests every year. In particular, *Helicoverpa armigera* (Hubner), *Scirpophaga incertulas* (Walker) and *Bactrocera oleae* pests are known to infect more than 150 crops across the world. These insects are particularly widespread in South Asia and the Mediterranean region. The female insects of these species release chemicals known as sex pheromones in the air for mating.

In this context, a group of scientists led by Prof. Santanu Bhattacharya, director, Indian Association for the Cultivation of Science (IACS), Kolkata, have developed a unique approach to detect the female sex pheromone of certain agriculturally hazardous pests and identify the prior infestation stage depending upon the pheromone concentration present in an agricultural field. Necessary action can then be taken to manage the pest infestation efficiently.



Fig. 1: Fabrication of SiO₂ based MEMS devices

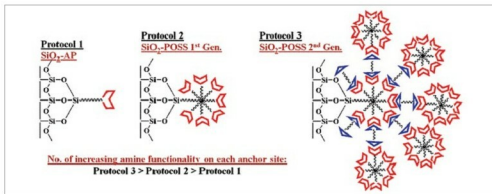


Fig. 2: Chemically functionalised MEMS devices for the selective detection of *Helicoverpa armigera*



Fig. 3: Covalently functionalised MEMS devices for the selective detection of *Bactrocera oleae*

A novel approach to detect chemical mating signals of hazardous crop pests, even at extremely low concentrations, thus prompting preventive measures to avoid crop losses